

Pebetsi Thokwane

Graduate Software Engineering Candidate | Physics PhD Candidate | Researcher

Cape Town, Western Cape, South Africa | +27 79 916 2461 | pebetsithokwane587@gmail.com
github.com/Pebetsit | LinkedIn

PROFESSIONAL PROFILE

Final-stage Physics PhD candidate with a strong background in scientific research, analytical problem-solving, experimental data interpretation and technical communication. Developing software engineering skills through practical web projects and self-directed learning in HTML, CSS, JavaScript, Python, Git and GitHub. Brings the persistence, structured thinking and attention to detail required to solve complex problems, learn unfamiliar technologies and contribute effectively within collaborative engineering teams.

SOFTWARE & TECHNICAL SKILLS

Web development HTML5, CSS3, JavaScript fundamentals, responsive web design	Programming & data Python, scientific data analysis, data visualisation, basic SQL
Tools Git, GitHub, VS Code, Microsoft Excel, Power BI, Tableau	Professional strengths Analytical thinking, problem-solving, technical writing, teamwork, communication, adaptability

SELECTED SOFTWARE PROJECTS

Scientific Calculator

Responsive browser-based calculator using HTML, CSS and JavaScript, with arithmetic operations, square root, square and cube functions.

Personal Portfolio Website

Multi-page portfolio presenting education, research, technical skills, software projects and a downloadable resume.

Pine City Zoo Website

Responsive multi-page HTML and CSS website with animal, attraction, map, restaurant, weather and feedback pages.

Superhero Website

Multi-page superhero-themed website with character profiles, image gallery, navigation, About and Contact pages.

Research Laboratory Website Redesign

Private multi-page research laboratory website independently designed using HTML and CSS.

EDUCATION

Doctor of Philosophy in Physics

University of South Africa | 2023 - Present
Thesis completed; awaiting formal submission and examination.

Master of Science in Physics

University of South Africa | 2021 - 2023

Bachelor of Science Honours in Physics

University of Venda | 2020

Bachelor of Science in Physics and Chemistry

University of Venda | 2017 - 2019
Best Physics Student award.

WORK EXPERIENCE

Laboratory Assistant - Physics Department

University of South Africa | 2022 - 2023

- Supported undergraduate laboratory practical sessions and research activities.
- Guided students in laboratory procedures, calculations, data analysis, interpretation and scientific report writing.
- Supported analytical instrumentation, equipment readiness, troubleshooting and safe laboratory operations.
- Collected, analysed and interpreted experimental data for materials-science research.

Physics Tutor

University of Venda | 2019 - 2021

- Provided academic support in undergraduate Physics, calculations and problem-solving.
- Assisted students with tutorials, assignments and examination preparation.

Laboratory Assistant / Demonstrator - Physics Department

University of Venda | 2019 - 2020

- Demonstrated undergraduate Physics experiments and guided students in data collection, error analysis and interpretation.
- Supported practical-session preparation, safe laboratory practice and scientific report writing.

RESEARCH EXPERIENCE

Doctoral research focuses on lanthanide-doped upconversion nanomaterials and their application in improving photon absorption in perovskite solar-cell materials. Experience includes solution combustion synthesis, experimental design, materials characterisation, data interpretation, scientific writing, technical presentations and manuscript preparation.

- Completed a PhD thesis in Physics.
- Published peer-reviewed research in materials science and renewable-energy-related nanomaterials.
- Prepared manuscripts, technical reports and responses to reviewer comments.
- Supported postgraduate laboratory research activities.

LABORATORY & TECHNICAL EXPERIENCE

Structural & optical characterisation Rigaku SmartLab XRD PerkinElmer Lambda 1050 UV-Visible Horiba Fluorolog-3 PL Horiba XploRA PLUS Raman Shimadzu IRTracer-100 FTIR	Thermal, luminescence & microscopy Risø DA-20 Thermoluminescence Thermogravimetric Analysis SEM sample preparation TEM
Research methods Materials synthesis Experimental design Scientific data acquisition Data interpretation	Laboratory practice Equipment troubleshooting Technical reporting Laboratory safety Student support

PUBLICATIONS & RESEARCH OUTPUTS

- Thokwane, P., Mbule, P. (2023). Analysis of the NaYF4:Yb3+, Er3+ nanocrystals: up-conversion luminescence, crystal structure and morphology influenced by the dopant concentration and annealing temperature. Bulletin of Materials Science.
- Mofokeng et al. (2024). Defects and Vacancies Effects on the Optoelectronic Properties of Two-Dimensional Nanomaterials. CRC Press.
- Thokwane, P. et al. (2025). Structural, optical and mechanical evaluation of solution-combusted NaYF4 as a host matrix for rare-earth doping: Experimental and DFT insights. Physica B: Condensed Matter.
- Thokwane, P. et al. (2026). Structural, Optical and Electronic Properties of Er3+- and Yb3+-Er3+-Doped Mixed-Phase NaYF4 Nanoparticles: Experimental and DFT Insights. Materials Chemistry and Physics.
- Sintwa, N., Talane, T., Thokwane, P., et al. Structural and Thermoluminescence Properties of ZnTiO3 Nanostructures Synthesized Using Hoodia Gordonii Ethanolic Extract: Kinetic Validation and Dosimetric Performance. SSRN preprint.

CERTIFICATIONS

- Google Data Analytics Professional Certificate - Coursera
- Applied Data Science Lab - WorldQuant University
- Introduction to Data Analysis Using Excel - Coursera
- Data Storytelling with Power BI - Coursera

REFERENCES

Available upon request.